

WASP-50b

R-1744

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_131029 · created 2026-08-02T18:57:02+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456594.869 +/- 0.09 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.051
Transit depth (Rp/Rs)^2	3.04 +/- 1.79 [%]
Orbital Inclination (inc)	89.39 +/- 2.88
Airmass coefficient 1 (a1)	1.15 +/- 0.1
Airmass coefficient 2 (a2)	-0.09 +/- 0.16
Scatter in the residuals of the lightcurve fit is	8.91 %
Best Comparison Star	None
Optimal Aperture	2.53
Optimal Annulus	4.5
Transit Duration (day)	0.048 +/- 0.037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.174 ± 0.051 vs 0.1404 (deviation 0.6σ)
Duration fit vs published	0.048 d vs 0.0754 d (deviation 0.67σ)
Mid-time O–C	82.2 min
Depth SNR	3.1
Residual scatter	8.91 %

Light curve

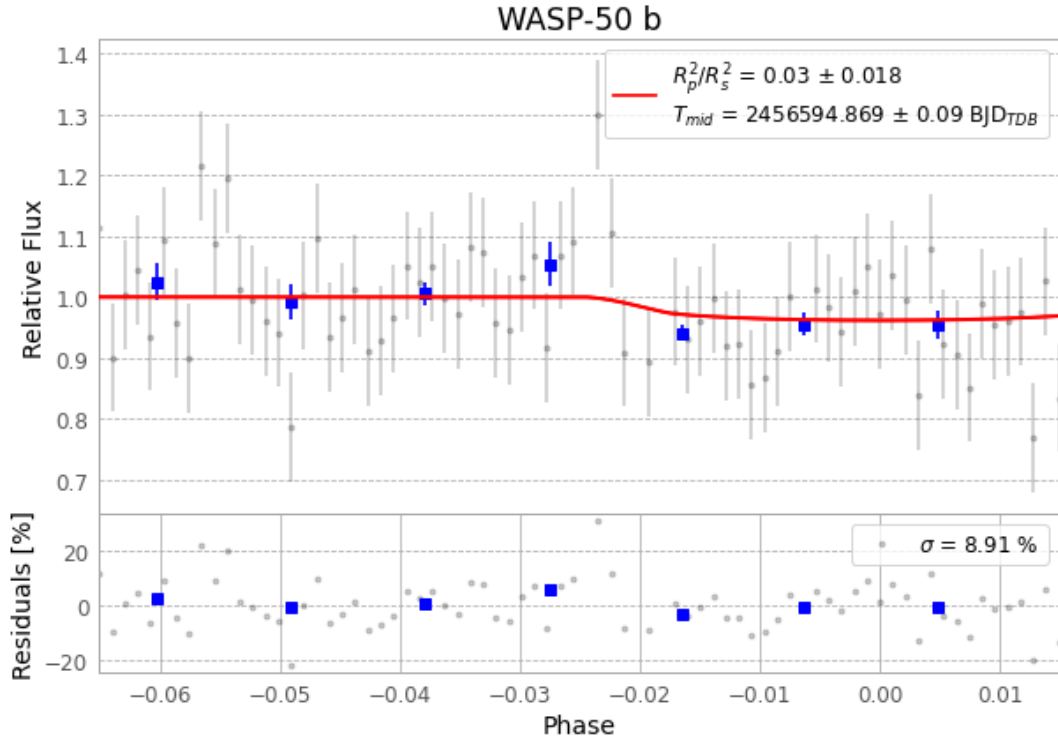


Figure 1 — FinalLightCurve_WASP-50 b_2013-10-28.png (SHA-256 1147c188021eac6e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [333, 397], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 545ae0a8e25c63fb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), WASP-50131029053912.FITS → WASP-50131029092412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-131029.log (4fc9a820c54c...), FinalLightCurve_WASP-50 b_2013-10-28.png (1147c188021e...), FinalLightCurve_WASP-50 b_2013-10-28.csv (ee7eef008702...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 18:57Z.